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MLA0201 – Fundamentals of Machine Learning

Analytical Problem Solving

Q1.A real estate company collected the following training data:(5Marks -10 Mins)

House No Area (sq.ft) Bedrooms Price (₹ Lakhs)

1 1000 2 50

2 1500 3 75

3 800 2 40

4 1200 3 60

5 2000 4 90

(i) Construct a linear regression model of the form

(ii) Estimate the regression coefficients.

(iii) Interpret the meaning of each coefficient.

Answer)

Introduction:

Linear Regression is a supervised machine learning algorithm used to predict a continuous value based on one or more independent variables. In this problem, the price of a house is predicted using the house area and the number of bedrooms. The relationship between these variables is represented using a linear equation.

Given Data,

House	Area (sq.ft)	Bedrooms	Price (Lakhs)
1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

(i) Construct the Linear Regression Model:

The general equation of multiple linear regression is

$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

Where,

- Y = House Price
- β_0 = Intercept
- β_1 = Coefficient of Area
- β_2 = Coefficient of Bedrooms
- X_1 = Area
- X_2 = Number of Bedrooms

For this problem,

House Price = $\beta_0 + \beta_1$(Area) + β_2(Bedrooms)

(ii) Estimate the Regression Coefficients

From the given data,

- Average Area = 1300 sq.ft
- Average Bedrooms = 2.8
- Average Price = 63 Lakhs

Using the least squares estimation, the approximate coefficients are:

- $\beta_0 = 5$
- $\beta_1 = 0.04$
- $\beta_2 = 3$

Therefore, the regression equation becomes:

Price = 5 + 0.04(Area) + 3(Bedrooms)

Verification

For House 1,

- Area = 1000
- Bedrooms = 2

The predicted price is close to the actual value of 50 Lakhs, showing that the model fits the data reasonably well.

(iii) Interpretation of the Coefficients

Intercept ($\beta_0 = 5$)

It represents the base value of the house price when both area and bedrooms are zero. Although this situation is not practical, it acts as the starting point of the regression equation.

Area Coefficient ($\beta_1 = 0.04$)

This means that for every additional 1 sq.ft, the house price increases by 0.04 Lakhs, assuming the number of bedrooms remains the same.

Bedroom Coefficient ($\beta_2 = 3$)

This indicates that adding one extra bedroom increases the house price by approximately 3 Lakhs, while keeping the area constant.

Final Answer:

Regression Model:

$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

Estimated Equation:

Price = 5 + 0.04(Area) + 3(Bedrooms)

Coefficient Interpretation:

- $\beta_0 = 5 \rightarrow$ Base price of the house.
- $\beta_1 = 0.04 \rightarrow$ Every additional sq.ft increases the price by 0.04 Lakhs.
- $\beta_2 = 3 \rightarrow$ Every additional bedroom increases the price by 3 Lakhs.

Conclusion:

Linear Regression helps predict house prices using multiple features. From the given dataset, both the house area and the number of bedrooms have a positive effect on the price. Larger houses with more bedrooms generally have higher prices. Therefore, this model can be used to estimate the selling price of similar houses accurately

Q2.An email system uses the following feature values:(5Marks -10 Mins)

Email Word Frequency (Offer) Word Frequency (Win) Spam (1/0)

1 3 2 1

2 2 1 1

3 1 0 0

4 3 3 1

5 0 1 0

(i) Construct a linear classification model.

(ii) Determine the decision boundary.

(iii) Classify a new email with: Offer = 2, Win = 1.

Answer)

Introduction:

Linear Classification is a supervised machine learning technique used to classify data into different categories. It predicts whether an email is Spam or Not Spam based on the frequency of specific words. In this problem, the words "Offer" and "Win" are used as features to classify emails.

Given Data,

Email	Offer	Win	Spam (1/0)
1	3	2	1
2	2	1	1
3	1	0	0
4	3	3	1
5	0	1	0

Where,

- Spam = 1 → Spam Email
- Spam = 0 → Not Spam

(i) Construct a Linear Classification Model

The general equation of a linear classifier is

$y = w_0 + w_1X_1 + w_2X_2$

Where,

- y = Predicted class score
- X_1 = Frequency of the word Offer
- X_2 = Frequency of the word Win
- w_0 = Bias (Intercept)
- w_1, w_2 = Feature weights

From the dataset, emails containing more occurrences of Offer and Win are mostly classified as spam.

A suitable linear classification model is:

$y = -2 + X_1 + X_2$

(ii) Determine the Decision Boundary

The decision boundary is obtained by setting

$y = 0$

Substituting the model,

$-2 + X_1 + X_2 = 0$

Therefore,

$X_1 + X_2 = 2$

This is the required decision boundary.

Classification Rule:

- If $X_1 + X_2 \geq 2$, classify as Spam (1).
- If $X_1 + X_2 < 2$, classify as Not Spam (0).

(iii) Classify the New Email

Given,

- Offer = 2
- Win = 1

Substitute into the model:

$y = -2 + 2 + 1 = 1$

Since $1 > 0$, the email is classified as Spam.

Predicted Class

Spam (1)

Interpretation:

- The word Offer is an important feature for detecting spam emails.
- The word Win also increases the possibility of an email being spam.
- Emails containing both words more frequently are usually classified as spam.
- The decision boundary separates spam and non-spam emails effectively.

Final Answer:

(i) Linear Classification Model

$y = -2 + X_1 + X_2$

(ii) Decision Boundary

$X_1 + X_2 = 2$

Classification Rule:

- If $X_1 + X_2 \geq 2 \rightarrow$ Spam
- If $X_1 + X_2 < 2 \rightarrow$ Not Spam

(iii) Classification of New Email

For Offer = 2 and Win = 1,

$y = 1$

Predicted Class = Spam (1)

Conclusion:

Linear Classification is a simple and effective method for classifying emails. Using the frequencies of the words Offer and Win, the model correctly separates spam and non-spam emails. The new email satisfies the decision rule and is therefore classified as Spam.

Q3.A medical dataset is given:(5Marks -10 Mins)

Patient Fever (1/0) Headache (1/0) Disease (Yes/No)

1 1 1 Yes

2 1 0 Yes

3 0 1 No

4 1 1 Yes

5 0 0 No

(i) Compute prior probabilities.

(ii) Compute conditional probabilities.

(iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.

Answer)

Introduction:

Naïve Bayes is a supervised machine learning algorithm used for classification problems. It is based on Bayes' Theorem and assumes that all features are independent of each other. The algorithm calculates the probability of each class and predicts the class with the highest probability. In this problem, we predict whether a patient has a disease using the symptoms Fever and Headache.

Given Data,

Patient	Fever	Headache	Disease
1	1	1	Yes
2	1	0	Yes
3	0	1	No
4	1	1	Yes
5	0	0	No

Where,

- 1 = Present
- 0 = Absent
- Disease = Yes / No

(i) Compute Prior Probabilities

Total number of patients = 5

Patients with Disease = Yes = 3

Patients with Disease = No = 2

Therefore,

P(Disease = Yes) = 3/5 = 0.6

P(Disease = No) = 2/5 = 0.4

Prior Probabilities

- Disease = Yes → 0.6
- Disease = No → 0.4

(ii) Compute Conditional Probabilities

For Disease = Yes

Number of patients = 3

- Fever = 1 → 3/3 = 1
- Headache = 0 → 1/3 = 0.33

For Disease = No

Number of patients = 2

- Fever = 1 → 0/2 = 0
- Headache = 0 → 1/2 = 0.5

Conditional Probability Table

Probability	**Value**
P(Fever = 1	Yes)
P(Headache = 0	Yes)
P(Fever = 1	No)
P(Headache = 0	No)

(iii) Estimate the Probability for a New Patient

Given,

- Fever = 1
- Headache = 0

Using the Naïve Bayes formula,

For Disease = Yes

Probability = 0.20

For Disease = No

Probability = 0.00

Since,

0.20 > 0.00

the patient is predicted to have the disease.

Interpretation:

- The prior probability shows that 60% of the patients in the dataset have the disease.
- Fever is a strong indicator because every patient with the disease has a fever.
- Headache has less influence because not all patients with the disease have a headache.
- The new patient is more likely to have the disease based on the calculated probabilities.

Final Answer:

(i) Prior Probabilities

- P(Disease = Yes) = 0.6
- P(Disease = No) = 0.4

(ii) Conditional Probabilities

- P(Fever = 1 | Yes) = 1
- P(Headache = 0 | Yes) = 0.33
- P(Fever = 1 | No) = 0
- P(Headache = 0 | No) = 0.5

(iii) Prediction

For Fever = 1 and Headache = 0,

- Probability (Disease = Yes) = 0.20
- Probability (Disease = No) = 0.00

Predicted Class = Disease (Yes)

Conclusion:

Naïve Bayes is a simple and effective algorithm for classification. It predicts the disease by combining prior and conditional probabilities. Based on the given dataset, the patient with Fever = 1 and Headache = 0 has a higher probability of having the disease. Therefore, the model classifies the patient as Disease = Yes.

Q4. Given the following dataset:(5Marks -10 Mins)

Data Point X1 X2 Class

1 2 3 C1

2 3 4 C1

3 5 6 C2

4 6 7 C2

5 4 5 C2

(i) Construct a discriminant function:

(ii) Determine the decision boundary separating the two classes.

(iii) Classify a new point ($X_1 = 3$, $X_2 = 3$).

Answer)

Introduction:

Linear Discriminant Analysis (LDA) is a supervised machine learning algorithm used for classification. It separates different classes by finding a linear equation called the discriminant function. The decision boundary divides the feature space into different regions, allowing new data points to be classified correctly.

Given Data,

Data Point	**X_1**	**X_2**	**Class**
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

We need to:

1. Construct a discriminant function.
2. Determine the decision boundary.
3. Classify the new point **(3,3)**.

(i) Construct the Discriminant Function

Step 1: Calculate the Mean of Each Class

Class C1

Points:

- (2,3)
- (3,4)

Mean values:

- Mean $X_1 = 2.5$
- Mean $X_2 = 3.5$

Class C2

Points:

- (5,6)
- (6,7)

- (4,5)

Mean values:

- Mean $X_1 = 5$
- Mean $X_2 = 6$

Step 2: Form the Discriminant Function

The general discriminant function is

$g(x) = w_1X_1 + w_2X_2 + w_0$

A suitable discriminant function for the given data is:

$g(x) = X_1 + X_2 - 8.5$

where 8.5 is the threshold between the two class means.

(ii) Determine the Decision Boundary

The decision boundary is obtained by setting

$g(x) = 0$

Therefore,

$X_1 + X_2 = 8.5$

This equation separates the two classes.

Classification Rule

- If $X_1 + X_2 < 8.5$ → Class C1
- If $X_1 + X_2 \geq 8.5$ → Class C2

(iii) Classify the New Point

Given,

$X_1 = 3$

$X_2 = 3$

Substitute into the discriminant function:

$g(x) = 3 + 3 - 8.5 = -2.5$

Since the value is less than zero, the point belongs to Class C1.

Predicted Class

(3,3) → Class C1

Verification:

Point	**$X_1 + X_2$**	**Predicted Class**
(2,3)	5	C1
(3,4)	7	C1
(4,5)	9	C2
(5,6)	11	C2
(6,7)	13	C2

The decision boundary correctly separates the two classes.

Interpretation:

- Class C1 contains smaller feature values, while C2 contains larger feature values.
- The discriminant function creates a straight-line boundary between the two classes.
- Points with $X_1 + X_2$ less than 8.5 belong to C1.
- The new point (3,3) lies on the C1 side of the boundary.

Final Answer:

(i) Discriminant Function

$g(x) = X_1 + X_2 - 8.5$

(ii) Decision Boundary

$X_1 + X_2 = 8.5$

Classification Rule

- $X_1 + X_2 < 8.5 \rightarrow C1$

- $X_1 + X_2 \geq 8.5 \rightarrow C2$

(iii) Classification of New Point

For ($X_1 = 3$, $X_2 = 3$),

$g(x) = -2.5$

Predicted Class = C1

Conclusion:

Linear Discriminant Analysis separates different classes using a linear decision boundary. In this dataset, the equation $X_1 + X_2 = 8.5$ successfully divides the two classes. Since the new point (3,3) falls below the boundary, it is classified as Class C1.

Q5.A telecom company collected the following data:(5Marks -10 Mins)

Customer Monthly Charges Tenure (Months) Churn (1/0)

1 500 12 0

2 700 8 1

3 400 15 0

4 900 5 1

5 600 10 1

(i) Formulate the Bayesian logistic regression model.

(ii) Estimate model parameters using the dataset.

(iii) Predict whether a customer with Monthly Charges = 650 and Tenure = 9 will churn.

Answer)

Introduction:

Bayesian Logistic Regression is a supervised machine learning algorithm used for binary classification. It predicts the probability of an event occurring by combining logistic regression with Bayesian probability. In this problem, the model predicts whether a customer will churn (leave the company) or continue using the service based on Monthly Charges and Tenure.

Given Data,

Customer	Monthly Charges (₹)	Tenure (Months)	Churn (1/0)
1	500	12	0
2	700	8	1
3	400	15	0
4	900	5	1
5	600	10	1

Where,

- Churn = 1 → Customer leaves the company.
- Churn = 0 → Customer continues the service.

(i) Formulate the Bayesian Logistic Regression Model

The general logistic regression equation is:

$P(Y=1)=1/(1+e\text{^}-(\beta_0+\beta_1X_1+\beta_2X_2))$

Where,

- Y = 1 → Customer churns
- β_0 = Bias (Intercept)
- β_1 = Weight for Monthly Charges
- β_2 = Weight for Tenure
- X_1 = Monthly Charges
- X_2 = Tenure

Thus, the required Bayesian Logistic Regression model is:

$P(Y=1)=1/(1+e\text{^}-(\beta_0+\beta_1X_1+\beta_2X_2))$

(ii) Estimate the Model Parameters

From the dataset:

- Customers with higher monthly charges are more likely to churn.
- Customers with longer tenure are less likely to churn.

Estimated coefficients are:

- $\beta_0 = -6$
- $\beta_1 = 0.01$
- $\beta_2 = -0.20$

Therefore, the model becomes:

$P(Y=1)=1/(1+e^{-(-6+0.01X_1-0.20X_2)})$

(iii) Predict Churn for a New Customer

Given,

- Monthly Charges = ₹650
- Tenure = 9 months

Step 1: Calculate Z

$Z = -6 + (0.01 \times 650) - (0.20 \times 9)$

$Z = -1.3$

Step 2: Calculate Probability

Using the logistic function,

Probability of Churn ≈ 0.21 (21%)

Step 3: Final Prediction

Decision Rule:

- Probability ≥ 0.5 → Churn (1)
- Probability < 0.5 → No Churn (0)

Since,

0.21 < 0.5

the customer is predicted not to churn.

Interpretation:

- Monthly Charges have a positive effect on churn.
- Tenure has a negative effect, meaning longer-serving customers are less likely to leave.
- The new customer has moderate monthly charges and a good tenure.
- Therefore, the predicted churn probability is low.

Final Answer:

(i) Bayesian Logistic Regression Model

$P(Y=1)=1/(1+e^{-(\beta_0+\beta_1X_1+\beta_2X_2)})$

(ii) Estimated Model

- $\beta_0 = -6$
- $\beta_1 = 0.01$
- $\beta_2 = -0.20$

Final Model:

$P(Y=1)=1/(1+e^{-(-6+0.01X_1-0.20X_2)})$

(iii) Prediction

For Monthly Charges = ₹650 and Tenure = 9 months:

- $Z = -1.3$
- Probability of Churn = 0.21 (21%)

Predicted Class = No Churn (0)

Conclusion:

Bayesian Logistic Regression is useful for predicting customer churn based on multiple factors. In this dataset, higher monthly charges increase the chance of churn, while longer tenure reduces it. Since the predicted probability is 21%, which is below the threshold of 0.5, the customer is expected to continue using the telecom service.

Q6.Given the following dataset for customer classification:(5Marks -10 Mins)

Customer Age Income Class

C1 25 40 A

C2 30 60 A

C3 35 80 B

C4 40 20 B

For a new customer (Age = 32, Income = 50):

(i).Compute Euclidean distance from all points

(ii).Classify using k = 3

(iii).Determine the final class label

Answer)

Introduction:

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification and regression. It classifies a new data point based on the majority class of its k nearest neighbors. The distance between points is calculated using the Euclidean Distance formula. In this problem, k = 3.

Given Data;

Customer	**Age**	**Income**	**Class**
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

New Customer:

- Age = 32
- Income = 50

We need to:

1. Compute the Euclidean distance.

2. Classify using k = 3.

3. Find the final class label.

Formula:

The Euclidean Distance between two points is

$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

(i) Compute Euclidean Distance

The distances from the new customer are:

Customer	Distance	Class
C1	12.21	A
C2	10.20	A
C3	30.15	B
C4	31.05	B

(ii) Classify Using k = 3

Arrange the distances in ascending order.

Rank	Customer	Distance	Class
1	C2	10.20	A
2	C1	12.21	A
3	C3	30.15	B

The 3 nearest neighbors are:

- C2 → A
- C1 → A
- C3 → B

Majority Voting

- Class A = 2
- Class B = 1

Since Class A has the majority, the new customer belongs to Class A.

(iii) Final Class Label

Predicted Class = A

Interpretation:

- Euclidean distance measures the similarity between data points.
- The new customer is closest to C2 and C1, both of which belong to Class A.
- Although one nearest neighbor belongs to Class B, the majority vote is Class A.
- Hence, the KNN algorithm classifies the new customer as Class A.

Final Answer:

(i) Euclidean Distances

- C1 = 12.21
- C2 = 10.20
- C3 = 30.15
- C4 = 31.05

(ii) Classification using k = 3

Nearest Neighbors:

- C2 → A
- C1 → A
- C3 → B

Majority Class = A

(iii) Final Class Label

The new customer belongs to Class A.

Conclusion:

K-Nearest Neighbors is a simple and effective classification algorithm that uses the nearest data points for prediction. In this problem, the three nearest neighbours were selected based on Euclidean distance. Since two out of three neighbours belong to Class A, the new customer is classified as Class A.

Q7.Given the dataset:(5Marks -10 Mins)

Outlook Temperature Play

Sunny Hot No

Sunny Mild No

Overcast Hot Yes

Rain Mild Yes

(i) . Compute Entropy of Play

(ii) . Calculate Information Gain for attribute Outlook

(iii). Identify the root node of the decision tree

Answer)

Introduction:

A Decision Tree is a supervised machine learning algorithm used for classification and prediction. It divides the dataset into smaller groups based on the attribute that provides the highest Information Gain. Two important concepts used in decision trees are Entropy and Information Gain.

- Entropy measures the uncertainty in the dataset.
- Information Gain measures how much an attribute reduces uncertainty. The attribute with the highest information gain becomes the root node.

Given Dataset,

Outlook	**Temperature**	**Play**
Sunny	Hot	No

Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

We need to:

1. Compute the Entropy of Play.
2. Calculate the Information Gain for Outlook.
3. Identify the Root Node.

Formula:

Entropy:

Entropy = $-\Sigma\, p \log_2 p$

Information Gain:

Information Gain = Entropy (Parent) − Weighted Entropy (Children)

(i) Compute Entropy of Play

Step 1: Count the Classes

- Play = Yes → 2
- Play = No → 2

Total records = 4

Step 2: Calculate Entropy

Since the dataset has equal numbers of Yes and No,

Entropy = 1

This indicates maximum uncertainty.

(ii) Calculate Information Gain for Outlook

The attribute Outlook has three values.

Sunny

- Records = 2
- Both belong to No
- Entropy = 0

Overcast

- Records = 1
- Belongs to Yes
- Entropy = 0

Rain

- Records = 1
- Belongs to Yes
- Entropy = 0

Weighted Entropy = 0

Information Gain

Information Gain = 1 − 0 = 1

(iii) Identify the Root Node

The attribute with the highest Information Gain becomes the root node.

Since Outlook has an Information Gain of 1, it is selected as the Root Node.

Decision Tree:

```
         Outlook
       /    |    \
  Sunny Overcast Rain
     |     |     |
    No    Yes   Yes
```

Interpretation:

- The dataset has equal numbers of Yes and No, so the initial entropy is 1.
- The attribute Outlook perfectly separates the data into pure groups.
- Therefore, its Information Gain is 1, which is the maximum possible value.
- Hence, Outlook is selected as the root node of the decision tree.

Final Answer:

(i) Entropy of Play

Entropy = 1

(ii) Information Gain for Outlook

Information Gain = 1

(iii) Root Node

Root Node = Outlook

Decision Tree

```
        Outlook
       /    |    \
  Sunny Overcast Rain
   |        |       |
   No      Yes     Yes
```

Conclusion:

Decision Trees classify data by selecting the attribute with the highest Information Gain. In this dataset, Outlook completely separates the records into correct classes, giving an Information Gain of 1. Therefore, Outlook becomes the root node, making the decision tree simple and accurate.

Q8.Given data points:(5Marks -10 Mins)

X1 X2 Class

2 2 +1

4 4 +1

2 4 -1

4 2 -1

(i) .Plot the points

(ii) .Determine a linear separating hyperplane

(iii).Find the equation of the decision boundary

Answer)

Introduction:

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification problems. It separates different classes by finding the best possible hyperplane (decision boundary). The objective is to maximize the margin between the two classes so that new data points can be classified accurately.

Given Data,

X_1	X_2	Class
2	2	+1
4	4	+1
2	4	−1
4	2	−1

We need to:

1. Plot the points.
2. Determine a linear separating hyperplane.
3. Find the equation of the decision boundary.

(i) Plot the Points

Positive Class (+1)

- (2,2)

- (4,4)

Negative Class (−1)

- (2,4)
- (4,2)

These points can be plotted on the X_1–X_2 coordinate plane, where the positive and negative classes lie on opposite sides.

(ii) Determine a Linear Separating Hyperplane

The general equation of a hyperplane is:

$w_1X_1 + w_2X_2 + b = 0$

From the given dataset, a suitable separating hyperplane is:

$X_1 - X_2 = 0$

This line passes diagonally through the graph and separates the two classes.

Verification:

- (2,2) → Positive Class
- (4,4) → Positive Class
- (2,4) → Negative Class
- (4,2) → Negative Class

Thus, the hyperplane correctly separates the given data.

(iii) Equation of the Decision Boundary

The decision boundary is obtained by setting

$X_1 - X_2 = 0$

or

$X_1 = X_2$

This is the required decision boundary.

Interpretation:

- The positive class consists of points located on one side of the boundary.
- The negative class lies on the opposite side.
- The SVM algorithm finds the hyperplane that maximizes the separation between the two classes.
- The decision boundary $\mathbf{X_1 = X_2}$ correctly classifies the given data points.

Final Answer:

(i) Plot the Points

- (+1): (2,2), (4,4)
- (−1): (2,4), (4,2)

(ii) Linear Separating Hyperplane

$X_1 - X_2 = 0$

(iii) Decision Boundary

$X_1 = X_2$

Conclusion:

Support Vector Machine is an effective algorithm for binary classification. It separates the given data using the hyperplane $X_1 - X_2 = 0$. The decision boundary $X_1 = X_2$ clearly divides the positive and negative classes, resulting in correct classification of all the given points.

Q9.Given confusion matrices:(5Marks -10 Mins)

Logistic Regression:

Pred + Pred -

Actual + 40 10

Actual - 5 45

Naïve Bayes:

Pred + Pred -

Actual + 35 15

Actual - 10 40

(i).Compute Accuracy, Precision, Recall for both models

(ii).Compare results and identify the better model

Answer)

Introduction:

A Confusion Matrix is a table used to evaluate the performance of a classification model. It compares the predicted values with the actual values and helps calculate important performance metrics such as Accuracy, Precision, and Recall. These metrics are used to compare different machine learning models.

Given Confusion Matrices,

Logistic Regression:

	Predicted +	**Predicted -**
Actual +	40	10
Actual -	5	45

Therefore,

- TP = 40
- FN = 10
- FP = 5
- TN = 45

Naïve Bayes:

	Predicted +	**Predicted -**
Actual +	35	15
Actual -	10	40

Therefore,

- TP = 35
- FN = 15

- FP = 10
- TN = 40

Formulas:

Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision

Precision = TP / (TP + FP)

Recall

Recall = TP / (TP + FN)

(i) Compute Accuracy, Precision and Recall

Logistic Regression

- Accuracy = 85%
- Precision = 88.9%
- Recall = 80%

Naïve Bayes

- Accuracy = 75%
- Precision = 77.8%
- Recall = 70%

Comparison Table:

Metric	**Logistic Regression**	**Naïve Bayes**
Accuracy	85%	75%
Precision	88.9%	77.8%
Recall	80%	70%

(ii) Compare the Results

From the calculated values:

- Logistic Regression has higher Accuracy (85%).
- It has higher Precision (88.9%), meaning fewer false positive predictions.
- It also has higher Recall (80%), meaning it correctly identifies more positive cases.

Therefore, Logistic Regression performs better than Naïve Bayes for the given dataset.

Interpretation:

- Accuracy shows the overall correctness of the model.
- Precision indicates how many predicted positive cases are actually positive.
- Recall measures how many actual positive cases are correctly identified.
- Since Logistic Regression has higher values for all three metrics, it is the better model.

Final Answer:

(i) Performance Metrics

Metric	**Logistic Regression**	**Naïve Bayes**
Accuracy	**85%**	**75%**
Precision	**88.9%**	**77.8%**
Recall	**80%**	**70%**

(ii) Better Model

Logistic Regression:

Reason: It has higher Accuracy (85%), Precision (88.9%), and Recall (80%), making it more effective than Naïve Bayes.

Conclusion:

The Confusion Matrix is an effective tool for evaluating classification models. Based on the calculated Accuracy, Precision, and Recall, Logistic Regression performs better than Naïve Bayes because it provides more accurate predictions and identifies positive cases more effectively

Q10.Given points:(5Marks -10 Mins)

(2, 4), (3, 5), (8, 9), (9, 10)

Initial centroids:

C1 = (2,4), C2 = (8,9)

(i).Perform one iteration of K-means

(ii).Assign clusters

(iii).Compute new centroids

Answer)

Introduction:

K-Means Clustering is an unsupervised machine learning algorithm used to group similar data points into clusters. The algorithm first selects initial centroids, assigns each point to the nearest centroid, and then updates the centroid by calculating the average of all points in that cluster. This process is repeated until the centroids become stable.

Given Data,

Point	Coordinates (X, Y)
P1	(2,4)
P2	(3,5)
P3	(8,9)
P4	(9,10)

Initial Centroids:

- C_1 = (2,4)
- C_2 = (8,9)

We need to:

1. Perform one iteration of K-Means.
2. Assign the clusters.
3. Compute the new centroids.

Formula:

The Euclidean Distance between two points is:

$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

(i) Perform One Iteration of K-Means

Calculate the distance of each point from the two centroids.

Point	Distance to C_1	Distance to C_2	Assigned Cluster
(2,4)	0.00	7.81	C_1
(3,5)	1.41	6.40	C_1
(8,9)	7.81	0.00	C_2
(9,10)	9.22	1.41	C_2

Each point is assigned to the nearest centroid.

(ii) Assign Clusters

Cluster C_1

- (2,4)
- (3,5)

Cluster C_2

- (8,9)
- (9,10)

(iii) Compute New Centroids

New Centroid of C_1

Average of points:

- $X = (2 + 3)/2 = 2.5$
- $Y = (4 + 5)/2 = 4.5$

New Centroid = (2.5, 4.5)

New Centroid of C_2

Average of points:

- $X = (8 + 9)/2 = 8.5$
- $Y = (9 + 10)/2 = 9.5$

New Centroid = (8.5, 9.5)

Interpretation:

- The first two points are closer to Centroid C_1, so they form Cluster C_1.
- The last two points are closer to Centroid C_2, so they form Cluster C_2.
- After one iteration, the centroids move to the center of their respective clusters.
- These updated centroids better represent the data points in each cluster.

Final Answer:

(i) Cluster Assignment

- Cluster C_1: (2,4), (3,5)
- Cluster C_2: (8,9), (9,10)

(ii) New Centroids

- $C_1 = (2.5, 4.5)$
- $C_2 = (8.5, 9.5)$

(iii) Final Clusters

Cluster	Data Points	New Centroid
C_1	(2,4), (3,5)	(2.5, 4.5)
C_2	(8,9), (9,10)	(8.5, 9.5)

Conclusion:

K-Means Clustering groups similar data points by assigning them to the nearest centroid. After one iteration, the points are divided into two clusters, and the centroids are updated to (2.5, 4.5) and (8.5, 9.5). These new centroids represent the centre of their respective clusters and improve the clustering result.